

CHAPTER 6: WAN TECHNOLOGIES

WAN Concepts and Devices

WAN concepts are all about where your responsibility ends and where the Service Provider's (ISP) responsibility begins.

1. DTE (Data Terminal Equipment)

The **DTE** is the device that sits at the edge of the customer's network. It is the "source" or the "destination" of the data. Its job is to turn the user's data into signals that can be sent over the WAN.

- **The Device:** Usually a **Router**.

2. DCE (Data Circuit-terminating Equipment)

The **DCE** sits between the DTE and the WAN cloud. Its primary job is to provide the **clocking signal** (the "beat") that tells the DTE how fast to send data. It converts the data into a format that the service provider's physical lines can handle.

- **The Device:** A **Modem** or a **CSU/DSU** (Channel Service Unit/Data Service Unit).
- **Key Rule for Students:** In a laboratory environment (like Packet Tracer), when you connect two routers via a Serial Cable, you must decide which one is the "DCE" and set the clock rate. If you don't set the clock rate on the DCE side, the line will stay "down."

3. The Demarcation Point (The "Demarc")

The **Demarcation Point** is the precise physical point where the ISP's network ends and the customer's private network begins. It is essentially a "boundary line."

- **The Location:** Usually a grey box on the outside of a building or a patch panel in a server room.

POINT-TO-POINT AND PACKET-SWITCHED NETWORKS

Building a Wide Area Network (WAN) is like designing a transportation system for your data. Depending on your budget and security needs, you might choose a private "toll road" (**Point-to-Point**) or a shared "public highway" (**Packet-Switched**).

To secure these roads, we use **VPNs**, which act like armored tunnels through the public internet.

1. Point-to-Point Networks

A Point-to-Point link (also known as a **Leased Line**) is a dedicated, private physical connection between two fixed locations. No one else's data travels on this line.

- **Pros:** Guaranteed speed (bandwidth) and high security.
- **Cons:** Very expensive and difficult to scale (you need a new physical wire for every new location).

2. Packet-Switched Networks

In a Packet-Switched network (like the Internet, Frame Relay, or MPLS), your data is broken into small "packets." These packets share the same wires with data from millions of other people.

- **Pros:** Much cheaper than Point-to-Point; easy to add new locations.
- **Cons:** Speed can drop if the "highway" gets crowded (congestion).

INTRODUCTION TO VPNS

A **Virtual Private Network (VPN)** uses the public internet to create a secure, encrypted "tunnel." It gives you the security of a private road while using the cheap price of the public highway.

A. Site-to-Site VPNs

A Site-to-Site VPN connects two entire networks together permanently. The routers at each location do all the work; the employees don't even know a VPN is running.

B. Remote Access VPNs (The Secure Key)

A Remote Access VPN connects a **single user** to a central network. The user must usually open a piece of software (like Cisco AnyConnect) and log in.

References:

- Cisco Networking Academy. Networking Concepts and Routing & Switching Modules. Cisco Systems, Inc.